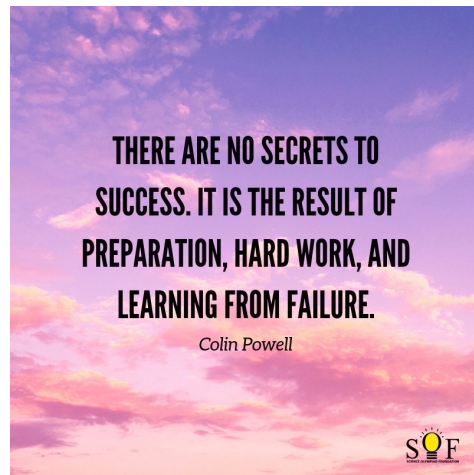


MODULUS & GCD



Good
Morning



Content

* Modulus

01. Modular Arithmetic

03. Fast pow function

04. Count pair with sum mod = 0

05. Intro to GCD

06. Properties of GCD



Modulus Operator

$A \% B$ = returns remainder when A is divided by B .

$$\text{Dividend} = \text{Divisor} * \text{quotient} + \text{remainder}$$

$$\text{rem} = \text{Dividend} - \underbrace{\text{Divisor} * \text{quotient}}$$

largest mul of divisor

$$\leq \text{dividend}$$

01. $10 \% 4 = 2$

$$= 10 - (\text{largest mul of } 4 \leq 10)$$

$$= 10 - 8 = 2$$

02. $13 \% 5 = 3$

$$= 13 - (\text{largest mul of } 5 \leq 13)$$

$$= 13 - 10 = 3$$

03. $-40 \% 7 = 2$

$$= -40 - (\text{largest mul of } 7 \leq -40)$$

$$= -40 - (-42) = 2$$

04. $-60 \% 9 = -60 - (-63)$

$$= -60 + 63 = 3$$

$$x \% M = \begin{matrix} \text{min} & \text{max} \\ \left[0 & M-1 \right] \end{matrix}$$

⇒ Modulus operator is used to limit/restrict the range of output.

$$\left. \begin{matrix} -\infty \\ \vdots \\ 0 \end{matrix} \right\} \% M = \{0 \text{ to } M-1\}$$

$M = 10^9 + 7 \rightarrow$ very close to int range
 $\searrow$ prime no.

* Modular Arithmetic (% with +, -, *, /)

01. $(a + b) \% M \Rightarrow (a \% M + b \% M) \% M$

$$\left. \begin{matrix} a = 13 \\ b = 9 \end{matrix} \right\} \text{max value i can store} = 15$$

$$m = 5$$

$$\underbrace{(13 + 9)}_{22} \% 5 = (13 \% 5 + 9 \% 5) \% 5$$

$$22 \% 5 = (3 + 4) \% 5$$

$$\Rightarrow 2 = 7 \% 5 \Rightarrow 2$$

$$02. (a * b) \% M \Rightarrow (a \% M * b \% M) \% M$$

$$\left. \begin{array}{l} a = 13 \\ b = 9 \end{array} \right\} \text{max value i can store} = 15$$

$$m = 15$$

$$(13 * 9) \% 15 = (13 \% 15 * 9 \% 15) \% 15$$

$$117 \% 15 \Rightarrow (3 * 9) \% 15$$

$$\Rightarrow 12 \Rightarrow \underbrace{27 \% 15 = 12}$$

Whenever we have to calculate %, we should first let it go beyond the range of M.

$$03. (a - b) \% M = (a \% M - b \% M + M) \% M$$

$$\left. \begin{array}{l} a = 13 \\ b = 9 \end{array} \right\} \text{max value i can store} = 15$$

$$m = 5$$

$$(13 - 9) \% 5 = (13 \% 5 - 9 \% 5) \% 5$$

$$= 4 \% 5 \Rightarrow (3 - 4) \% 5$$

$$\Rightarrow 4 \Rightarrow -1 \% 5 \Rightarrow -1$$

$$\Rightarrow \underbrace{-1 + 5}_{+M} = \boxed{4}$$

Note → Either add +M at last if answer is negative.

OR

Add it in the formula as $(a \% M - b \% M + M)$

$$04. (a/b) \% M \longrightarrow (a * b^{-1}) \% M \longrightarrow \underbrace{\text{Inverse modulo}}_{\text{Fermat theorem}}$$

$$05. (((a \% M) \% M) \% M) \longrightarrow a \% M$$

$$06. (a^b \% M) \longrightarrow ((a \% M)^b \% M)$$

Evaluate : $(37^{103} - 1) \% 12$

$$\Rightarrow (37^{103} \% 12 - 1 \% 12 + 12) \% 12$$

$$\Rightarrow ((37 \% 12)^{103} \% 12 - 1 + 12) \% 12$$

$$\Rightarrow ((1)^{103} \% 12 + 11) \% 12$$

$$\Rightarrow (1 + 11) \% 12$$

$$\Rightarrow 12 \% 12 \Rightarrow 0$$

Fast Power with mod using recursion

Problem Statement

Given three integers a, n and m. Find $a^n \% m$ using recursion.

Constraints:

$$1 \leq a \leq 10^9$$

$$1 \leq n \leq 10^9$$

$$2 \leq m \leq 10^9$$

Idea 1 $\rightarrow$ Calculate a^n first & then take $\%M$

$$(10^9)^{10^9}$$

X

Approach 2 $\rightarrow$ Use fast pow function

$$2^6 \rightarrow 2^3 * 2^3 \rightarrow a^n = a^{n/2} * a^{n/2} \quad (n \text{ is even})$$

$$2^9 \rightarrow 2^4 * 2^4 * 2 \rightarrow a^n = a^{n/2} * a^{n/2} * a \quad (n \text{ is odd})$$

```
int powmod (int a, int n, int m) {
```

```
    if (n == 0) return 1;
```

```
    intlong y = powmod (a, n/2, m);
```

```
    if (n % 2 == 0) return (y * y) % m;
```

```
    else return (y * y * a) % m;
```

```
}
```

X

Will this
work? $\rightarrow$ No

```
intlong y = powmod (a, n/2, m);
```

// mod y = 10^9

```
if (n % 2 == 0) return (y * y) % m
```

// mod ans = 10^9

$$\underbrace{10^9 * 10^9}_{10^{18}}$$

$$\text{return } (y * y * a) \% M$$

$\underbrace{y * y}_j * \overset{\downarrow}{k}$

$$(j * k) \% M = (j \% M + k \% M) \% M$$

$$((y * y) \% M * a \% M) \% M$$

$\underbrace{(10^9 * 10^9) \% M}_{10^{18} \% M}$

```

int powmod (int a, int n, int m) {
    if (n == 0) return 1;

    long y = powmod (a, n/2, m);

    if (n % 2 == 0) return (int)(y * y) % M;

    else
        return (int)((y * y) % M * a % M) % M;
}

```

8:04 → 8:13 AM

* Given N array elements, find count of pairs such that $(A[i] + A[j]) \% m == 0$

Note :- $i \neq j$ & $\text{pair}(i, j) = \text{pair}(j, i)$

$A = \{ 4 \quad 7 \quad 6 \quad 5 \quad 5 \quad 3 \}$ $m = 3 \rightarrow \text{Ans} = 5$
0 1 2 3 4 5

i	j	A[i]	A[j]	$(A[i] + A[j]) \% m$
0	3	4	5	$(4 + 5) \% 3 = 0$
0	4	4	5	$(4 + 5) \% 3 = 0$
1	3	7	5	$(7 + 5) \% 3 = 0$
1	4	7	5	$(7 + 5) \% 3 = 0$
2	5	6	3	$(6 + 3) \% 3 = 0$

Brute force $\rightarrow$ Consider all pairs & get the sum of each pair & take modulus.

if (modulus answer == 0) count++;

Tc : $O(N^2)$

Sc : $O(1)$

Idea 2

$$(A[i] + A[j]) \% M = 0$$

$\Downarrow$

$$(\underbrace{A[i] \% M} + \underbrace{A[j] \% M}) \% M = 0$$

Edge case $\rightarrow 0 + 0$

$$1 + (M-1)$$

$$2 + (M-2)$$

$$3 + (M-3)$$

$\vdots$

$$K + (M-K)$$

$$A[] = \{ 13, 14, 22, 3, 32, 19, 16 \} \quad M = 4$$

$$\text{rem}[] = \{ 1, 2, 2, 3, 0, 3, 0 \}$$

Ans = 4

$m = 5$

$$A[] = \{ \underset{0}{6}, \underset{1}{7}, \underset{2}{5}, \underset{3}{11}, \underset{4}{19}, \underset{5}{20}, \underset{6}{9}, \underset{7}{15}, \underset{8}{14}, \underset{9}{13}, \underset{10}{12}, \underset{11}{23} \}$$

$$\text{rem} = \{ 1, 2, 0, 1, 4, 0, 4, 0, 4, 3, 2, 3 \}$$

$$\text{count}[s] = \left\{ \begin{array}{ccccc} 3 & 2 & 2 & 2 & 3 \\ 0 & 1 & 2 & 3 & 4 \end{array} \right\}$$

$\uparrow$

$\uparrow$

pair of remainder

$$\text{cnt}[1] * \text{cnt}[5-1] = 2 * 3 = 6$$

$$\text{cnt}[2] * \text{cnt}[5-2] = 2 * 2 = 4$$

$$\text{cnt}[0] \& \text{cnt}[0] = {}^3C_2 = 3$$

$$\frac{3 * (3-1)}{2}$$

13 pairs

```
int countpairs (int[] A, int M)
```

```
int[] cnt = new int [M];
```

```
for (i=0; i<n; i++)
```

```
    int rem = A[i] % M
```

```
    cnt[rem] ++;
```

```
ans = 0
```

```
int x = cnt[0]
```

```
ans = ans + x * (x-1) / 2;
```

```
if (m % 2 == 0)
```

```
    int y = cnt[m/2];
```

```
    ans = ans + y * (y-1) / 2;
```

```
i = 1, j = M-1
```

```
while (i < j)
```

```
    ans = ans + (cnt[i] * cnt[j])
```

```
    i++; j--;
```

```
return ans;
```

TC: $O(N+M)$

SC: $O(M)$

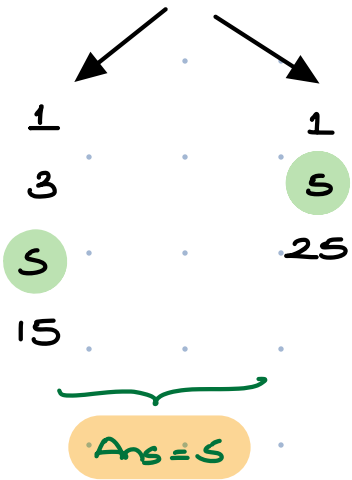
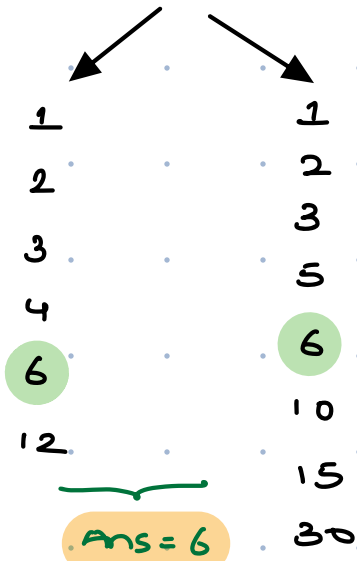
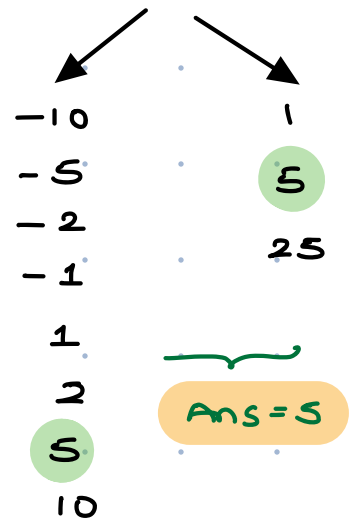
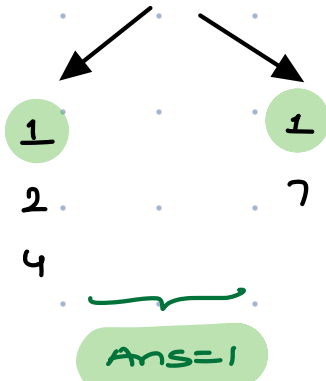
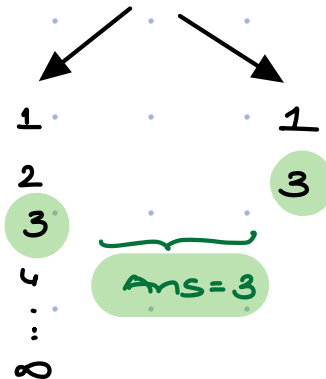
G, CD.

GCD = Greatest common Divisor

HCF = Highest common factor

$\text{GCD}(A, B) = \text{largest no. that divides both } A \text{ \& } B.$

• GCD (15, 25)


$$\text{GCD}(12, 30)$$

$$\text{GCD}(-10, 25)$$

$$\text{GCD}(4, 7)$$
 $C, CD(0, 3)$ 

* Properties of GCD

01. $\text{GCD}(A, B) = \text{GCD}(B, A)$

02. $\text{GCD}(A, B) = \text{GCD}(|A|, |B|)$

03. $\text{GCD}(0, A) = |A|$

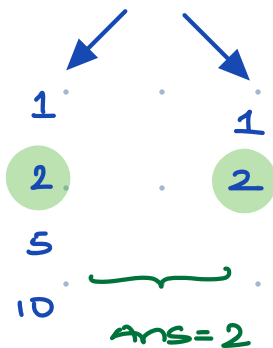
04. $\text{GCD}(A, B, C) = \text{GCD}(A, \text{GCD}(B, C))$

$= \text{GCD}(\text{GCD}(A, B), C)$

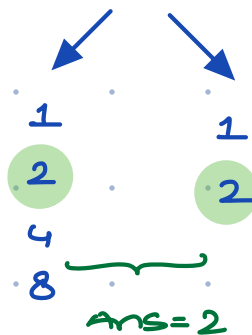
$= \text{GCD}(B, \text{GCD}(A, C))$

05. $\text{GCD}(A, B) = \text{GCD}(A - B, B)$

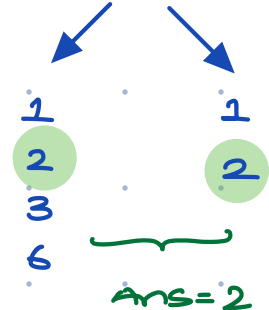
$\text{GCD}(10, 2)$



$\text{GCD}(8, 2)$



$\text{GCD}(6, 2)$



$\text{GCD}(A, B) = \text{GCD}(A - B, B)$

$= \text{GCD}(A - 2B, B)$

$\vdots$

$= \text{GCD}(A - xB, B)$

$A - \text{greatest mul of } B \leq A$

$$\text{GCD}(A, B) = \text{GCD}(A \% B, B)$$

* Write a function for GCD(A, B)

01. $\text{GCD}(24, 16) \rightarrow \text{GCD}(8, 16) \rightarrow \text{GCD}(8, 16) \rightarrow \text{infinite loop}$

$$\text{GCD}(A, B) = \text{GCD}(B, A \% B)$$

01. $\text{GCD}(24, 16) \rightarrow \text{GCD}(16, 8) \rightarrow \text{GCD}(8, 0) \quad \underline{\underline{\text{Ans} = 8}}$

02. $\text{GCD}(14, 21) \rightarrow \text{GCD}(21, 14) \rightarrow \text{GCD}(14, 7) \rightarrow \text{GCD}(7, 0) \quad \underline{\underline{\text{Ans} = 7}}$

03. $\text{GCD}(105, 10) \rightarrow \text{GCD}(10, 5) \rightarrow \text{GCD}(5, 0) \quad \underline{\underline{\text{Ans} = 5}}$

04. $\text{GCD}(23, 25) = \text{GCD}(25, 23) \rightarrow \text{GCD}(23, 2) \rightarrow \text{GCD}(2, 1)$

Ans = 1

$\downarrow$
 $\text{GCD}(1, 0)$

```
int GCD(A, B)
{
    if (B == 0) return A;
    return GCD(B, A % B)
}
```

TC: $O(\log(\max(A, B)))$

SC: $O(\log(\max(A, B)))$

